

WHARTEXA™

Human Umbilical Cord MSC-Derived Extracellular Vesicles

20 & 60 Billion Bioactive Particles | Frozen Sterile Suspension | 5 mL

Human umbilical cord-derived exosomes are considered a first-class exosome source due to their high biologic activity, superior regenerative potential, low immunogenicity, and rich concentration of growth factors essential for advanced tissue repair and cellular renewal. They are obtained from young mesenchymal stem cell sources, which provides stronger biologic signaling and consistent purity. Their advanced regenerative profile makes them highly valued in aesthetic, dermatologic, and cellular recovery applications.





HUMAN UMBILICAL CORD DRIVEN EXOSOMES

Derived from Wharton's Jelly of the umbilical cord, rich in growth factors and cytokines.

-  High Regenerative Potential
Supports tissue regeneration and healing
-  Supports Immune Modulation
Helps balance immune response
-  Enhances Tissue Repair
Promotes faster and effective recovery
-  Low Immunogenicity & Safe
High biocompatibility with minimal risk
-  High Purity & Potency
Ensures consistent and reliable results



HUMAN ADIPOSE TISSUE DRIVEN EXOSOMES

Sourced from human adipose tissue, abundant in healing and anti-inflammatory factors.

-  Supports Skin Rejuvenation
Improves skin texture and elasticity
-  Reduces Inflammation & Swelling
Promotes healing in inflamed tissues
-  Promotes Angiogenesis & Healing
Supports new blood vessel formation
-  High Yield & Consistency
Reliable quality and scalable production
-  Clinically Relevant Results
Effective in aesthetic and therapeutic applications

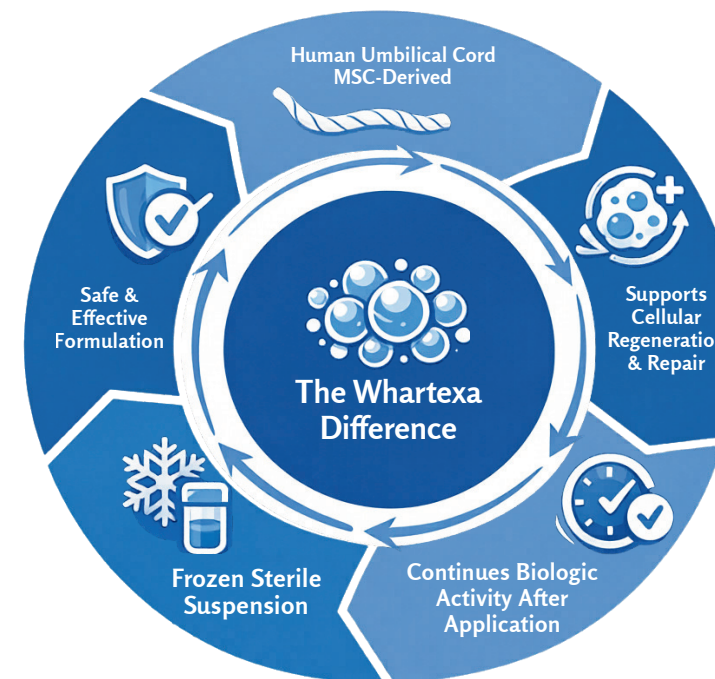


FIBROBLASTS EXOSOMES

Fibroblast-derived exosomes rich in growth factors and regenerative signals that support skin renewal and repair.

-  Boosts Collagen & Elastin Production
Enhances skin firmness, elasticity and youthfulness
-  Supports Skin Regeneration
Promotes faster tissue repair and cell renewal
-  Reduces Fine Lines & Wrinkles
Improves skin texture and reduces signs of aging
-  High Purity & Potency
High-quality exosomes for optimal biological activity
-  Safe & Biocompatible
Minimal immunogenicity, safe for clinical and aesthetic use

The Science Behind WHARTEXA™



-  **Human Umbilical Cord MSC-Derived**
Formulation utilizes purified extracellular vesicles derived from human umbilical cord mesenchymal stem cells (MSCs).
-  **Supports Cellular Regeneration & Repair**
Promotes cellular regeneration, tissue repair, and overall hair follicle rejuvenation.
-  **Continues Biologic Activity After Application**
Extracellular vesicles retain intact biologic components that remain active and functional even after application.

Manufactured by Nanoventra

-  **Precision Exosome Manufacturing**
Controlled scientific production focused on high-purity extracellular vesicle solutions.
-  **Advanced Purification**
Tangential flow filtration (TFF) and size-exclusion chromatography for vesicle integrity.
-  **Quality Validation**
Nanosight analysis, fluorescent staining, Sterility, endotoxin, and particle concentration testing.

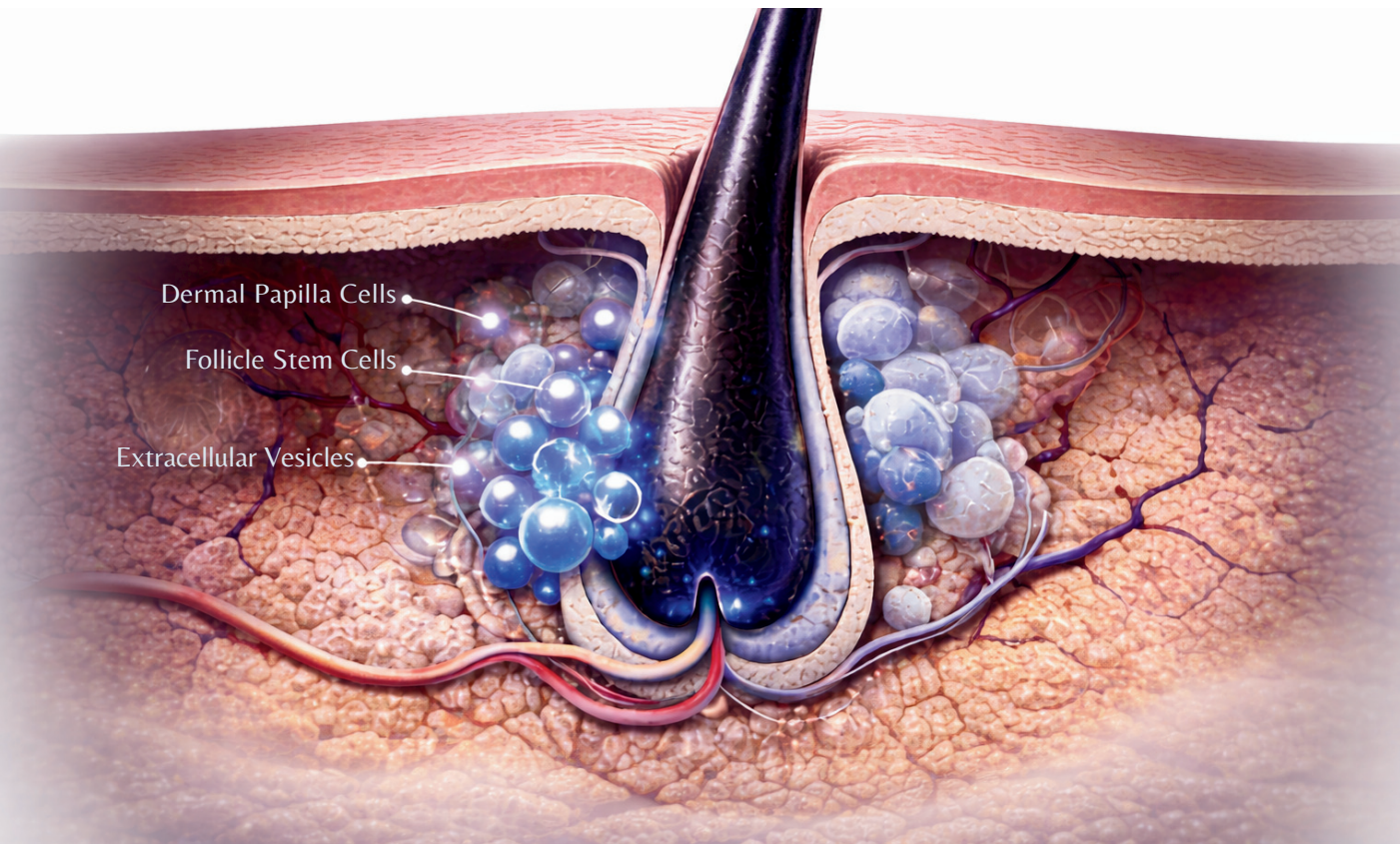


Nanoventra



HUMAN EXOSOME TECHNOLOGY

FOR ADVANCED SCALP BIOLOGIC SUPPORT



BIOLOGIC FORMULATION

- Umbilical Cord MSC-Derived Exosomes
- Sterile Frozen Suspension
- NO Peptides or Cosmetic Additives
- NO Synthetic Growth Factors

SCIENTIFIC PROFILE

- miRNAs & Growth Factors
- Cytokine Signals
- Native Vesicle Integrity

TARGETED INTERACTION

- Dermal Papilla Microenvironment
- Follicular microenvironment support
- Scalp Tissue Signaling

PROFESSIONAL APPLICATIONS

- Post-Microneedling
- Scalp Mesotherapy
- Physician-Controlled Delivery

About the Formulation

Derived from Human Umbilical Cord MSCs for Professional Use

COMPOSITION PROFILE

- Human Umbilical Cord MSC-Derived Exosomes
- Frozen Liquid Suspension Format
- Sterile, Ready-to-Use Format
- Without hyaluronic acid or added peptide/protein fillers
- Without artificial growth factors or cosmetic thickening agents

BIOLOGICAL POSITIONING

- Exosomes are extracellular vesicles naturally contain miRNAs, cytokine-associated signaling fractions, membrane proteins, and growth-related signaling molecules.
- HUC-MSC-derived exosomes are biologic extracellular vesicles evolved within the Human body to support regulatory signals at different levels of the hair follicle structure through directed vesicular transfer. Exosome-based signaling is believed to result in optimal regenerative support.



VERIFIED
FROZEN
STERILE

KEY BENEFITS

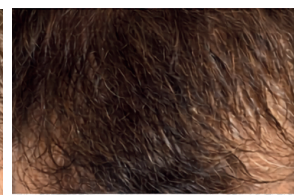
- Focuses on supporting dermal papilla cell microenvironment
- Targets the perifollicular pathway following controlled micro-injury
- Contains naturally occurring signaling molecules and microRNAs
- Maintains biologic exosome purity without adjunct peptide or cosmetic fillers

RESULTS

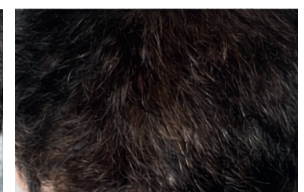
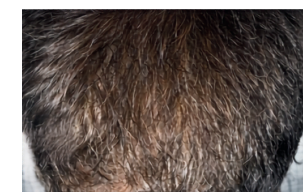
Before



After



Scalp Response After Two Sequential Exosome Treatments



Hair Density Progression Following Controlled Exosome Protocol

